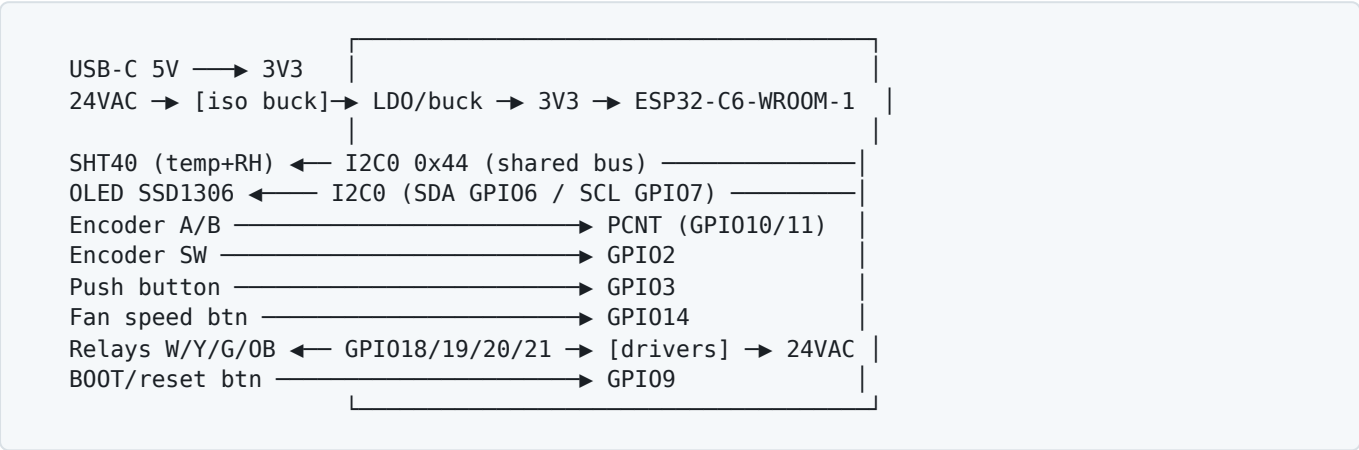


Hardware Design

Covers the block diagram, **pin assignment**, the SHT40 room sensor, input/output stages, power, and the BOM. Machine-readable copies live in `hardware/pinout.csv` and `hardware/bom.csv`.

1. Block diagram



2. Pin assignment (ESP32-C6-WROOM-1)

The ESP32-C6 GPIO matrix is flexible; the assignment below avoids the **strapping pins** (GPIO4, 5, 8, 9, 15), the **USB-Serial-JTAG** pins (GPIO12/13), the **UART0 console** (GPIO16/17), and the internal SPI-flash pins (GPIO24-30).

Signal	GPIO	On-chip peripheral	Direction	Notes
I²C SDA (OLED + SHT40)	GPIO6	I2C0	I/O	4.7 kΩ pull-up to 3V3; shared bus
I²C SCL (OLED + SHT40)	GPIO7	I2C0	O	4.7 kΩ pull-up to 3V3; shared bus
Encoder A / CLK	GPIO10	PCNT ch0	IN	Hardware quadrature decode
Encoder B / DT	GPIO11	PCNT ch0	IN	Hardware quadrature decode
Encoder switch	GPIO2	GPIO (ISR)	IN	Internal pull-up; press = select
Push button	GPIO3	GPIO (ISR)	IN	Internal pull-up; mode / back
Fan-speed button	GPIO14	GPIO (ISR)	IN	Internal pull-up; cycles Auto/Low/Med/High (−1 = unused)
Relay W (heat)	GPIO18	GPIO	OUT	Active-high to driver
Relay Y (cool/compressor)	GPIO19	GPIO	OUT	Active-high to driver
Relay G (fan enable)	GPIO20	GPIO	OUT	Active-high; any fan speed > 0
Relay O-B (reversing valve)	GPIO21	GPIO	OUT	Heat-pump only
Fan taps G_LOW/MED/HIGH (opt.)	−1	GPIO	OUT	Multi-speed blower; one-hot; disabled (−1) by default
Occupancy / PIR (opt.)	−1	GPIO	IN	Motion input; disabled (−1) = manual Home/Away only
Factory-reset button	GPIO9*	GPIO	IN	Re-uses BOOT (strapping, pulled-up)
Console UART TX/RX	GPIO16/17	UART0	—	Debug/log; keep free
USB D−/D+	GPIO12/13	USB-Serial-JTAG	—	Flash/monitor; keep free

* GPIO9 is a strapping pin, used here only for the BOOT button (externally pulled to the level the bootloader expects). Do not repurpose it for a peripheral that drives it at boot.

These names are mirrored 1:1 in `firmware/main/Kconfig.projbuild` so every pin is a menuconfig option — change the board without touching code.

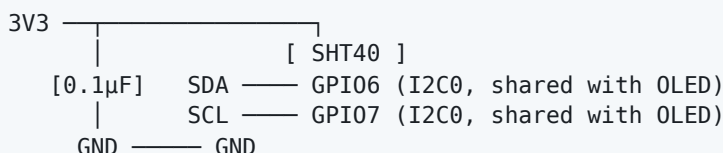
3. Room sensor (SHT40)

The room sensor is a **Sensirion SHT40** — a digital, factory-calibrated temperature + relative-humidity sensor on I²C. It is the only room sensor: there is no analog front-end, no ADC, and no thermistor.

Part	Interface	Provides	Address	Notes
SHT40 (SHT40-AD1B)	I ² C	temperature + relative humidity	0x44 (0x45 for -BD1B)	shares the OLED bus; no extra MCU pins

3.1 Wiring

The SHT40 is a 4-pin part (VDD / GND / SDA / SCL) that sits on the **same I²C0 bus** as the OLED (SDA GPIO6, SCL GPIO7, shared 4.7 kΩ pull-ups). The firmware creates one I²C master bus and adds both the OLED (0x3C) and the SHT40 (0x44) to it, so adding the sensor costs **no extra GPIO**. Decouple VDD with 100 nF close to the part.



3.2 Reading & conversion

The `sht4x` component issues the **high-precision measure** command (`0xFD`), reads 6 bytes (temperature word + CRC, humidity word + CRC), validates both bytes with the Sensirion **CRC-8** (poly 0x31, init 0xFF), and converts the raw ticks:

$$\begin{aligned}
 T \text{ [}^\circ\text{C]} &= -45 + 175 \cdot S_T / 65535 \\
 RH \text{ [\%]} &= -6 + 125 \cdot S_RH / 65535 \quad (\text{clamped to } 0\ldots100 \text{ \%})
 \end{aligned}$$

A user **calibration offset** (`offset` , 0.01 °C) is added after conversion. These pure functions are host-unit-tested (`firmware/test/host/test_sht4x.c`), including the datasheet CRC vector `0xBEEF` → `0x92` .

3.3 Accuracy & fault handling

- **Datasheet accuracy:** ±0.2 °C (typ.) temperature, ±1.8 %RH (typ.) — comfortably within the ±0.3 °C comfort-band target (NFR-1), with no board-level calibration required.
- **Fault:** a failed read or CRC mismatch ⇒ report fault, blank the Matter `LocalTemperature` , and force all relays off (FR-12).
- **Self-heating:** negligible at the ~1 Hz sampling used here (single-shot high-precision reads, sensor idle between samples).

4. OLED display

- **Default:** SSD1306, 128×64, monochrome, I²C, address **0x3C** (0x3D selectable).
- **Alternates:** SH1106 128×64 (firmware flag), SSD1306 128×32 0.91" (reduced layout).
- Shared **I²C0 bus @ 400 kHz** — the OLED (0x3C) and the optional **SHT40** (0x44) sit on the same two wires. The firmware creates one I²C master bus and adds both devices to it.

OLED options

Part	Size	Interface	Controller	Use here
0.96" mono OLED	128×64	I ² C	SSD1306	default
1.3" mono OLED	128×64	I ² C	SH1106	firmware flag
0.91" mono OLED	128×32	I ² C	SSD1306	compact layout
0.95" color OLED	96×64	SPI	SSD1331	<i>only if color required</i> — needs SPI pins (SCK/MOSI/DC/CS/RST), not I ² C

5. Inputs

- **Rotary encoder** (EC11-class): mechanical, ~20 detents/rev, quadrature A/B. Decoded in hardware by the **PCNT** peripheral (glitch filter enabled), so no steps are missed even under load. A/B get RC debounce (10 nF + internal pull-ups) in addition to the PCNT glitch filter.
- **Encoder switch (SW)** and **push button**: momentary, active-low with internal pull-ups; debounced in firmware (`components/button`) with short/long-press detection.
- **Fan-speed button (optional)**: a dedicated momentary button on `CONFIG_THERMO_PIN_BTN_FAN` (default GPIO14, active-low, internal pull-up). Each short press cycles the fan speed **Auto** → **Low** → **Med** → **High** → **Auto**; the new speed shows in the OLED status bar and mirrors to the Matter Fan Control cluster. Leave the pin at −1 to use only the settings menu / Matter for fan control.
- **BOOT/reset button**: held ≥ **10 s** it opens an on-screen chooser — **Pairing** (re-open the Matter commissioning window) or **Factory reset** — rather than resetting immediately.
- **Occupancy / PIR sensor (optional)**: a digital motion output (e.g. HC-SR501, AM312, or a PIR module) on `CONFIG_THERMO_PIN_OCCUPANCY`. Active-high by default (idle low, high on motion); the firmware enables the opposite internal pull so an unconnected pin reads "no motion". Powered from 3.3 V (AM312) or 5 V (HC-SR501 — level-check the output). Leave the pin at −1 to use only the manual Home/Away toggle.

6. Outputs (HVAC)

- 4 channels: **W** (heat), **Y** (cool/compressor), **G** (fan), **O·B** (reversing).
- Each channel: MCU GPIO → driver → load. Options:
- **Mechanical relay** (SPST, 24 VAC/2 A) via NPN/MOSFET + flyback diode, or a ULN2003-style array. Add an RC snubber across contacts for inductive 24 VAC loads.
- **Solid-state relay / triac** (e.g. opto-triac + MOC302x) for silent, long-life switching of 24 VAC — **recommended**, with opto-isolation from logic.
- **Isolation**: keep the 24 VAC domain optically isolated from the 3.3 V logic domain.
- **Fail-safe**: drivers are active-high; at reset/brown-out GPIOs are high-Z and the loads are **off**. Firmware also forces off on fault (FR-12).

Fan speed (Low / Med / High / Auto)

The fan speed is a first-class control (settings menu + Matter Fan Control cluster). How it reaches the blower depends on the wiring, all firmware-selectable:

- **Single-speed (default):** just the `G` relay. Any non-Auto speed = fan on; `Auto` runs the fan only during a heat/cool call.
- **Multi-speed blower:** wire the optional `G_LOW` / `G_MED` / `G_HIGH` tap relays (Kconfig pins, default `-1` = unused). The firmware energizes exactly one tap for the active level (one-hot) plus `G` as a general enable.
- **ECM / 0-10 V / PWM (future hook):** map the level to a duty cycle on a spare LEDC pin; the level→output mapping lives in `components/relays`.

During a heat/cool call the fan runs at the configured **call speed** (`THERMO_FAN_CALL_SPEED` , default High). A fixed Low/Med/High selection circulates continuously even when idle.

7. Status indication (on the OLED — no LED)

There is **no status LED**. The always-on OLED already conveys every state, so a discrete RGB LED would be redundant hardware; each state is shown on the display:

State	On-screen indication
Uncommissioned / pairing	PAIRING screen (QR box + manual code)
Connected	filled link dot in the status bar (hollow ring when not)
Idle (no call)	bottom-left running-state text: IDLE
Calling for heat	HEATING + a filled dot in the setpoint pill; ▲ mode marker
Calling for cool	COOLING + a filled dot in the setpoint pill; ▼ mode marker
Sensor fault	full-screen FAULT screen (outputs forced off)
Identify (Matter)	a centered IDENTIFY banner over the current screen

8. Power

- **Bench:** USB-C 5 V → 3.3 V buck/LDO for the module. Simplest for development.
- **Field (24 VAC HVAC):** isolated 24 VAC → 5 V converter (or C-wire), then 3.3 V rail. Add bulk capacitance to ride out radio TX current peaks (Thread/BLE bursts). A "C wire" (common) is required for continuous power; power-stealing is out of scope for v1.
- Decoupling per ESP32-C6-WROOM-1 datasheet; keep the antenna keep-out clear.

9. PCB / enclosure notes

- Keep the module antenna at a board edge with the vendor keep-out; no copper/metal near it.
- Keep the SHT40 I²C traces short; route them away from switching nodes.
- Physically and electrically separate the 24 VAC section (creepage/clearance) from logic.

- Mount the SHT40 away from the MCU/relays' self-heating and with airflow to the room — ideally vented at the enclosure edge, or on a short remote pigtail off the shared I²C bus.

10. Bill of materials

Summary — full list in [hardware/bom.csv](#) :

Qty	Item	Notes
1	ESP32-C6-WROOM-1(U) module, 8 MB	Thread + BLE
1	SSD1306 128×64 I ² C OLED	0x3C
1	EC11 rotary encoder w/ switch	A/B/SW
1	Momentary push button	mode/back
1	Momentary push button	fan-speed cycle (Auto/Low/Med/High)
1	Sensirion SHT40 (SHT40-AD1B)	room sensor: temp + humidity, I ² C 0x44
2	4.7 kΩ resistor	I ² C pull-ups
4	Relay or opto-triac + driver	W/Y/G/O·B
4	Flyback diode / snubber	per relay
1	Power supply (USB-C and/or 24 VAC→5 V iso)	
—	Decoupling caps, connectors, terminal blocks	